

Database Management Systems

CPS 3740

Lecture 13

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Outline

- Continue last lecture
- Chapter 7
 - How to use the integrity enhancement feature in the CREATE TABLE statements.
- CREATE and keys
- Project and homework status

Team study research

- Use keanWise as an example
- Form the team. Must be 4 people
 - Penalty for 1 or 2 people
- Decide the case
- Identify the GUI
- Draw E-R diagram
- Convert to MySQL create statements
- Identify the function and required SQL operations
- **Team presentation in class**
- **No late submission**

Chapter 7

SQL – Data Definition
Focus on Table Creation

Chapter 7 - Objectives

- **Data types** supported by SQL standard.
- Purpose of **integrity** enhancement feature of SQL.
- How to define **integrity constraints** using SQL.
- How to use the integrity enhancement feature in the **CREATE** and **ALTER TABLE** statements.
- **View**

Data Types

- The basic data types available for attributes :
 1. Numeric:
 - integer numbers of various sizes — INT(n), SMALLINT(n)....
 - floating-point (real) numbers of various precision — FLOAT(n,d), DOUBLE(n,d).....
 2. Boolean: TRUE or FALSE
 3. Character string
 - fixed length—CHAR(n)
 - varying length— VARCHAR(n)
 4. Date & Time: DATE(), DATETIME(), TIME().....
 5. Blob (Binary large object)

The following employee are in the database.

ID	Login	Password	Name	Role
1	tiger	test	Victor Smith	M
2	panda	test	Joe Lee	E

MySQL Data Types

- <http://www.tutorialspoint.com/mysql/mysql-data-types.htm>
- Numeric Data Types:
 - INT, TINYINT, SMALLINT, MEDIUMINT, BIGINT
 - FLOAT(M,D), DOUBLE(M,D), DECIMAL(M,D)
- Date and Time Types:
 - DATE, DATETIME, TIMESTAMP, TIME, YEAR(M)
- String Types:
 - CHAR(M), VARCHAR(M),
 - BLOB or TEXT, (TINY, MEDIUM, LONG)
 - ENUM

Data size and range are very critical to database design

- <https://dev.mysql.com/doc/refman/8.0/en/integer-types.html>
- $2^{10} = 1024$, ~1 K
- $2^{20} = 1024 \times 1024$, ~1 Million
- $2^{30} = 1024 \times 1024 \times 1024$, ~1 Billion (or 1G)
- For 1B records, an extra byte will cost 1GB size (storage and memory)

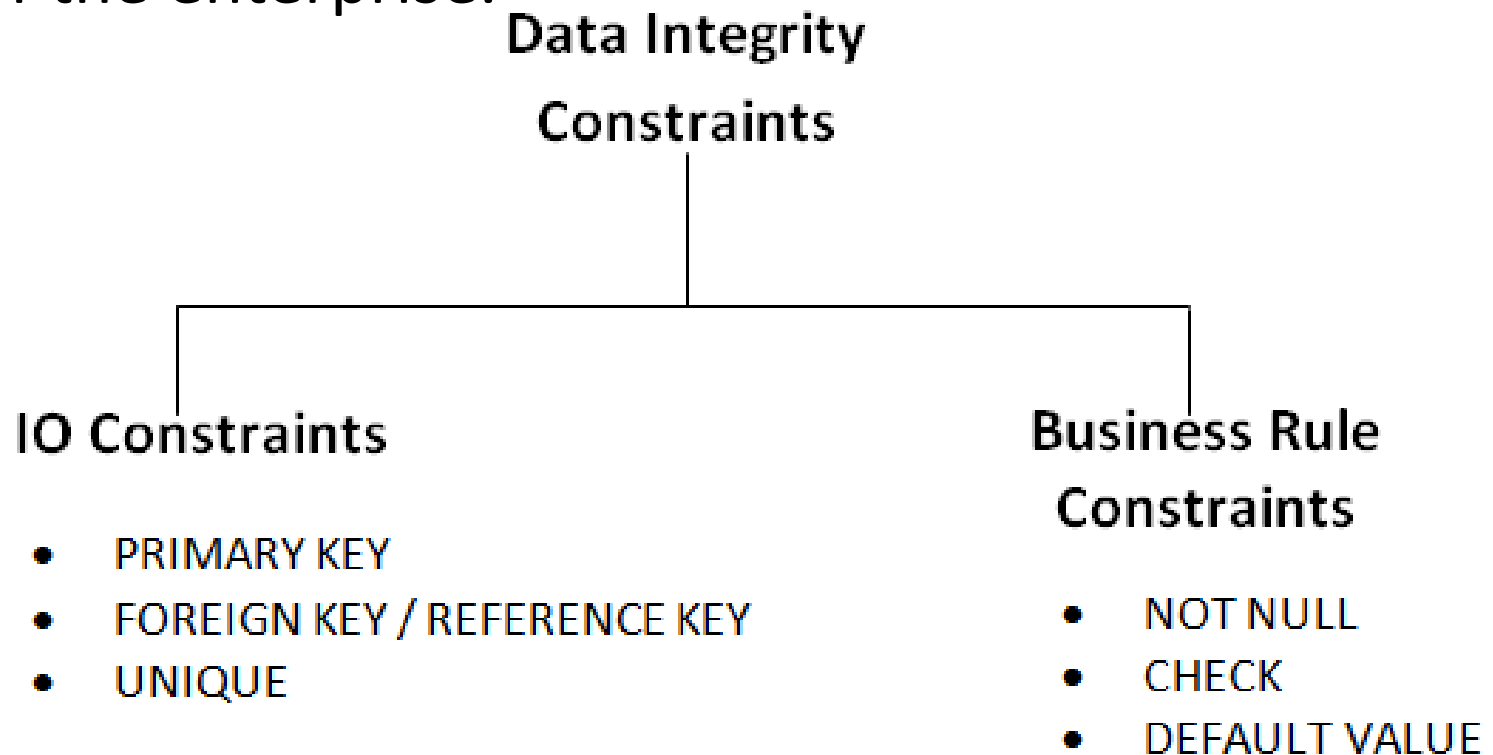
Type	Storage (Bytes)	Minimum Value Signed	Minimum Value Unsigned	Maximum Value Signed	Maximum Value Unsigned
TINYINT	1	-128	0	127	255
SMALLINT	2	-32768	0	32767	65535
MEDIUMINT	3	-8388608	0	8388607	16777215
INT	4	-2147483648	0	2147483647	4294967295
BIGINT	8	-2^{63}	0	$2^{63}-1$	$2^{64}-1$

Integrity Enhancement Feature

- Consider five **types** of integrity constraints:
 - required data
 - domain constraints
 - entity integrity
 - referential integrity
 - general constraints.

Integrity Constraints

- General Constraints
 - Additional rules specified by users or database administrators that define or constrain some aspect of the enterprise.



Integrity Constraints

- Entity Integrity
 - In a base relation, no attribute of a primary key can be null.
- Referential Integrity
 - If foreign key exists in a relation, either foreign key value must match a candidate key value of some tuple in its home relation or foreign key value must be wholly null.
 - Need to add values in a primary key before adding the same values in a foreign key
 - Need to delete values in foreign key before deleting the same values in a primary key
 - More discussions later

Primary Key Primary Key ?



StudentId	firstName	lastName	courseId
L0002345	Jim	Black	C002
L0001254	James	Harradine	A004
L0002349	Amanda	Holland	C002
L0001198	Simon	McCloud	S042
L0023487	Peter	Murray	P301
L0018453	Anne	Norris	S042

How About These Tables?

Suit	Value	No. times played
Hearts	Ace	5
Diamonds	Three	2
Hearts	Jack	3
Clubs	Three	5
Spades	Five	1

One is not enough to identify, but both combined make a unique value

Composite Keys

- A **simple key** is one that has only one attribute.
- A **composite key** contains two or more attributes.
- A primary key can be a composite key.

Buildings

B_ID	B_Name	Room#	Location	Type	Capacity
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Courses

Year	Term	Major	Course#	Section	Instructor	B_ID
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Integrity Enhancement Feature

Required Data

position VARCHAR(10) **NOT NULL**

Domain Constraints

(a) **CHECK**

sex CHAR NOT NULL
CHECK (sex IN ('M', 'F'))

- **MySQL does NOT support CHECK**

Integrity Enhancement Feature

(b) CREATE DOMAIN

```
CREATE DOMAIN DomainName [AS] dataType  
[DEFAULT defaultOption]  
[CHECK (searchCondition)]
```

For example:

```
CREATE DOMAIN SexType AS CHAR  
CHECK (VALUE IN ('M', 'F'));  
sex    SexType    NOT NULL
```

- **MySQL doesn't support user-defined domains**

Integrity Enhancement Feature

- *searchCondition* can involve a table lookup:

```
CREATE DOMAIN BranchNo AS CHAR(4)  
CHECK (VALUE IN (SELECT branchNo  
                    FROM Branch));
```

- Domains can be removed using **DROP DOMAIN**:

```
DROP DOMAIN DomainName  
[RESTRICT | CASCADE]
```

Integrity Enhancement Feature (IEF)

- Entity Integrity

- **Primary key** of a table must contain a **unique, non-null** value for each row.
- ISO standard supports **FOREIGN KEY** clause in CREATE and ALTER TABLE statements:

PRIMARY KEY(staffNo)

PRIMARY KEY(clientNo, propertyNo)

- Can only have **one** PRIMARY KEY clause per table. Can still ensure uniqueness for alternate keys using **UNIQUE**:

UNIQUE(telNo)

Examples in MySQL

```
create table Hotel(  
    hotelno varchar(10) primary key ,  
    hotelname varchar(20),  
    city varchar(20) );
```

```
create table Room(  
    roomno numeric(5),  
    hotelno varchar(10),  
    type varchar(10),  
    price decimal(5,2),  
    primary key (roomno, hotelno) );
```

General Constraints

- Could use CHECK/UNIQUE in CREATE and ALTER TABLE.
- Similar to the CHECK clause, also have:

**CREATE ASSERTION AssertionName
CHECK (searchCondition)**

MySQL doesn't support assertions

CHECK constraints are not supported by MySQL. You can define them, but they do nothing (as of MySQL 5.7)

General Constraints

```
CREATE ASSERTION StaffNotHandlingTooMuch  
CHECK (NOT EXISTS (SELECT staffNo  
                    FROM PropertyForRent  
                    GROUP BY staffNo  
                    HAVING COUNT(*) > 100))
```

Data Definition

- SQL DDL allows database objects such as schemas, domains, tables, views, and indexes to be created and destroyed.
- Main SQL DDL statements are:

CREATE SCHEMA

DROP SCHEMA

Same as **Create database** in MySQL

CREATE/ALTER DOMAIN

DROP DOMAIN

CREATE/ALTER TABLE

DROP TABLE

CREATE VIEW

DROP VIEW

- Many DBMSs also provide:

CREATE INDEX

DROP INDEX

Data Definition

- Relations and other database objects exist in an *environment*.
- Each environment contains one or more *catalogs*, and each catalog consists of set of schemas.
- Schema is named collection of related database objects.
- Objects in a schema can be tables, views, domains, assertions, collations, translations, and character sets. All have same owner.

CREATE SCHEMA

CREATE SCHEMA [Name |

AUTHORIZATION CreatorId]

DROP SCHEMA Name [RESTRICT | CASCADE]

- **With RESTRICT (default), schema must be empty or operation fails.**
- **With CASCADE, operation cascades to drop all objects associated with schema in order defined above. If any of these operations fail, DROP SCHEMA fails.**

**CREATE SCHEMA is a synonym
for CREATE DATABASE**

CREATE TABLE

- Creates a table with one or more columns of the specified *dataType*.
- With NOT NULL, system rejects any attempt to insert a null in the column.
- Can specify a DEFAULT value for the column.
- Primary keys should always be specified as NOT NULL.
- FOREIGN KEY clause specifies FK along with the referential action.

CREATE TABLE

```
CREATE TABLE TableName
{(colName dataType [NOT NULL] [UNIQUE]
[DEFAULT defaultOption]
[CHECK searchCondition] [,...]}
[PRIMARY KEY (listOfColumns),]
{[UNIQUE (listOfColumns),] [...,]}
{[FOREIGN KEY (listOfFKColumns)
REFERENCES ParentTableName [(listOfCKColumns)],
[ON UPDATE referentialAction]
[ON DELETE referentialAction ]] [,...]}
{[CHECK (searchCondition)] [,...] }
```

Example 7.1 - CREATE TABLE

```
CREATE DOMAIN OwnerNumber AS VARCHAR(5)  
    CHECK (VALUE IN (SELECT ownerNo FROM  
        PrivateOwner));  
  
CREATE DOMAIN StaffNumber AS VARCHAR(5)  
    CHECK (VALUE IN (SELECT staffNo FROM Staff));  
  
CREATE DOMAIN PNumber AS VARCHAR(5);  
  
CREATE DOMAIN PRooms AS SMALLINT;  
    CHECK(VALUE BETWEEN 1 AND 15);  
  
CREATE DOMAIN PRent AS DECIMAL(6,2)  
    CHECK(VALUE BETWEEN 0 AND 9999.99);
```

Example 7.1 - CREATE TABLE

```
CREATE TABLE PropertyForRent (  
    propertyNo  PNumber NOT NULL, ....  
    rooms      PRooms  NOT NULL DEFAULT 4,  
    rent       PRent    NOT NULL,  DEFAULT 600,  
    ownerNo    OwnerNumber NOT NULL,  
    staffNo     StaffNumber  
                Constraint StaffNotHandlingTooMuch ....  
    branchNo   BranchNumber NOT NULL,  
    PRIMARY KEY (propertyNo),  
    FOREIGN KEY (staffNo) REFERENCES Staff  
                ON DELETE SET NULL ON UPDATE CASCADE ....);
```

Auto increment

- The column value will be inserted by DBMS
- The value will be automatically increased by DMBS
- The column must be a key
- Example: (XXXX is your database login ID)

```
CREATE TABLE Test_XXXX (  
    id int NOT NULL auto_increment primary key,  
    name varchar (100)  
);
```
- Exercises: How to insert records?

MySQL auto_increment exercise

- Create table Test_xxxx (id int auto_increment primary key, name varchar(255), salary int);
- Insert two records
 - INSERT INTO Test_xxxx (name) values ('Austin'), ('Mary');
- Select the records
 - What happens to the salary column
- Change Austin and Marys' salaries
- Can we insert a record with id?
 - INSERT INTO Test_xxxx values (5, 'Austin', 50000);
- What happens to the id? What is the next value?

Project: PHP mysqli_insert_id()

- Get the **order_id (auto_increment)** for an **insert** statement

```
$query="INSERT INTO Orders_demo VALUES  
($customer_id,CURDATE());"
```

```
$result = mysqli_query($con,$query);
```

```
$order_id=mysqli_insert_id(); // fix it, google this function in w3schools
```

```
if ($order_id==0) { echo "Insert error! $query<br>\n"; }
```

Table information - Metadata

- Data and table information stored separately
- Repository of information (metadata) describing the data in the database.
- One of the fundamental components of DBMS.
- Metadata typically stores:
 - names, types, and sizes of data items;
 - constraints on the data;
 - names of authorized users;
 - data items accessible by a user and the type of access;
 - usage statistics.

DROP TABLE

DROP TABLE TableName [RESTRICT | CASCADE]

e.g. **DROP TABLE PropertyForRent;**

- **Removes named table and all rows within it.**
- **With RESTRICT, if any other objects depend for their existence on continued existence of this table, SQL does not allow request.**
- **With CASCADE, SQL drops all dependent objects (and objects dependent on these objects).**

Class exercise – create table

- Use the **class** database (eg. CPS3740_2024F)
- Follow steps by steps
 1. On web page, select option #2, get **CREATE** privilege
 2. Click “Execute the option” button
 3. Login to your **mysql** account
 4. Create the **Students_xxxx** table
 5. On web page, select option #6.3
 6. Select **Students_xxxx** table
 7. Click “Execute the option” button

Class exercise –delete & create/copy

- Create your own Staff_xxxx and copy all the records from dreamhome.Staff at the same time
`Create table Staff_xxxx as select * from dreamhome.Staff;`
- Write a SQL query to delete Ann's record from your own Staff_xxxx table.
- Delete all the records in your Staff_xxxx table.
- Remove your own Staff_xxxx table from the DB.

CREATE TABLE AS Statement – part 1

- You can also use the SQL CREATE TABLE AS statement to create a table from an existing table by copying the existing table's columns.
- It is important to note that when creating a table in this way, the new table will be populated with the records from the existing table (based on the SELECT Statement).
- It is considered to “COPY” a table
- Syntax:
CREATE TABLE new_table
AS (SELECT * FROM old_table);
- https://www.techonthenet.com/sql/tables/create_table2.php

CREATE TABLE AS Statement – part 2

- Copy selected columns from **another table**.

Syntax:

```
CREATE TABLE new_table  
AS (SELECT column_1, column2, ... column_n  
    FROM old_table);
```

- Copy selected columns from **multiple tables**.

Syntax:

```
CREATE TABLE new_table  
AS (SELECT column_1, column2, ... column_n  
    FROM old_table_1, old_table_2, ... old_table_n);
```

- https://www.techonthenet.com/sql/tables/create_table2.php

CREATE TABLE AS Statement – part 3

- Create a SQL table from another table without copying any values from the old table. Syntax:

```
CREATE TABLE new_table  
AS (SELECT * FROM old_table WHERE 1=2);
```

- https://www.techonthenet.com/sql/tables/create_table2.php
- **Important notes:**
 - Copying will copy these constraints:
 - Data type, size, null constraint, default value
 - Copying will NOT copy these constraints:
 - Primary key, foreign key, unique keys
- Be careful about the foreign keys.
 - We will discuss it later.

Class exercise - CREATE TABLE AS

- Copy dreamhome.Staff table to your own table Staff_xxxx under the class database.
- Do the following
 - > desc dreamhome.Staff;
 - > desc Staff_xxxx;
- Compare the difference

Home exercise – create your own project table

- Follow the requirements
 - All the functions should be implemented in PHP.
- Create your own table based on requirement of Homework
 - With a primary key and a foreign key.
 - With proper data types and constraints.
- Manually add a record into the table
- Login page
 - Enter user login ID and password in html.
 - Receive and authenticate (against the table based on project requirement 1).
 - Use POST method.
 - Show corresponding error messages if user login fails.
- Logout page
 - Clean and expire cookie.

Project – add records into your own table

- Only if user login successfully. Otherwise, ask user to login first.
 - Set Cookie.
 - Display the user information.
 - Show add link and search function. (ref: project requirement 3)
 - Add insert page
 - Valid the inputs based on the requirements.
 - No duplicated product name is allowed.
 - Add the inputs into your own table only all inputs are validated.
 - Show the corresponding error message when user clicks the add button.
 - Use POST method.

Project – part 3

- Only if user login successfully. Otherwise, ask user to login first.
 - search function
 - Use GET method.
 - Show the result in HTML <TABLE> that pattern match the keyword entered by the user and also show the number of matched records under the table.
 - The default text display color is black.
 - If the keyword is *, display all the product information and the number of products.
 - Based on the requirement, highlight some info in red or blue.
 - If no records matches the keyword, display the error messages base on the requirement.

Reading Assignment

- 7.1 Data type
- 7.2 Integrity Enhancement Feature
- 7.3 DDL – CREATE, DROP

Conclusion

- Do exercises in Chapter 6 and 7
- Work on project and homework